Yash Mahadev Parab



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Portfolio | GitHub

Engineering Management Graduate (RWTH Aachen) with core competencies in market analysis, corporate strategy, project management & entrepreneurial finance. Leveraging business acumen with data analysis & AI-driven automation to build-deploy digital management solutions and maximize ROI.

Key Skills

Strategy Development | International Market Analysis | Business Development | Project Management | Data Analysis | Stakeholder Collaboration | KPI Monitoring & Reporting | Process Optimization | Entrepreneurial Finance

Professional Experience

Market Research and Business Analyst Feb 2025 – Sep 2026
Fraunhofer FHR (Industrielle Hochfrequenzsysteme [IHS] Operations) *Wachtberg, Germany*

- Delivered end-to-end 25+ market intelligence reports across international high-tech sectors (New Space Tech, Military Drones, Electronic Warfare, Urban Mobility, AI-Intralogistics) to synthesize insights into actionable strategy recommendations.
- Informed business development decisions via 100+ competitor and product profiles, delivering strategic reports and executive presentations from secondary databases (Yole, Frost&Sullivan).
- Uncovered technology trends, innovation gaps and key players from 100M+ patents and publications by using Python and R based data analysis.
- Cut manual scouting time by ~90% and identified ~700 target opportunities from ~3K events for the team's marketing and networking plan, by automating data monitoring via a custom Python web scraper.

Research Assistant – Sustainability Analyst and Business Development Aug 2023 – Dec 2023
Production Engineering of E-Mobility Components (PEM RWTH) *Aachen, Germany*

- Built and presented interactive Power BI dashboards to stakeholders, validating circular-economy business models for 2nd-life batteries.
- Deployed a guided EU Reporting tool for CSRD scope + double materiality with EU Taxonomy checker and a spend-based Scope 3 Screening Estimator tool that ranks emission drivers according to EPA v1.4 factors.
- Identified 8-9% potential energy savings across 12+ e-mobility OEMs by designing and performing competitor analysis and financial assessment framework.
- Delivered audit-ready ESG reporting and strategy by owning the emissions data evaluation process through the full EV battery value chain.

Project Engineer – Process Automation and Optimization Jun 2020 – Sep 2022
R.K. Control Instruments Pvt. Ltd. *Thane, India*

- Strengthened project cost control and margin visibility by preparing cost estimates and quotations, tracking project costs, budgets and margins, and supporting production and capacity planning.
- Improved cross-functional collaboration and process efficiency by designing standardized analytical frameworks adopted team-wide.
- Led a key automation and optimization project by building an Excel-Python solution for 500+ mechanical engineering data files, that cut cycle time from 15 mins to 4 mins (~75%), directly enhancing business decisions and operational cost management.

Master's Thesis

Typologies of Business Angels: Classification Criteria and Differences in Investment Behavior
Live App: BA3F (Business Angels Funding Fit Finder) *(Note: app sleeps when idle)* Jul 2025 – Mar 2026
Innovation and Entrepreneurship Chair (TIME Research Area) *Aachen, Germany*

- Developed a classification framework for the informal venture capital market through structured analysis across 5 databases.
- Identified 5 typological dimensions – (demographics, experience, psychology, styles, organization) and mapped behavioral differences in risk tolerance, syndication and investment preferences.
- Operationalized framework into a live LLM-powered classifier (BA3F) that reads a public investor bio and returns a typology match plus pitch-tailoring guidance, reducing founder guesswork on who, how, and what to pitch.
- Addressed information asymmetry and angel-market heterogeneity for entrepreneurs and policymakers by delivering evidential outputs translated from deterministic findings in the exhaustive literature review.

Projects

Earnings Reaction Predictor (Personal): Equity Earnings-Surprise Analytics
Live App: ERP (Earnings Reaction Predictor) *(Note: app sleeps when idle, kindly wait ~10 seconds)*

- Built and deployed a dual-model ML pipeline forecasting the direction and magnitude of post-earnings stock moves across 403 earnings events, 51 US companies and 10 sectors.
- Validated on 100% out-of-time 2026 data (pre-2026 train, unseen 2026 test), delivering an interactive SHAP scenario-builder exposing the drivers behind each forecast across 7 features.

BCG Strategy Consulting: Telecom Company
- Projected ~4% profitability growth, ~2% revenue increase and improved ARPU for a telecom handset-leasing offer via market modelling of comparables across US, Singapore and Korea.

Mercedes Benz: In-Car Navigation Strategy
- Estimated a $160-$250M annual revenue opportunity by building a data-driven pricing model for proprietary in-car navigation.
- Mapped competitive landscape of premium Infotainment / Navigation services via Porter's forces and bundling as a growth lever for 33% profitability improvement.

Sony (Academic): Product Launch & Pricing
- Informed a B2B2C product launch business model by building a consumer and market trend analysis from 3 years of fashion retail data.
- Projected €1.6B in 5-year revenue by designing a value-based subscription pricing strategy.

PWC Power BI: Telecom Company
- Informed customer retention HR strategies by designing Power BI dashboards for churn and diversity KPIs by analyzing 5K+ customer survey responses.

Network Analysis & NLP (Personal): Witcher Books Project
- Visualized an interactive Pyvis network and computed Louvain communities to show characters' relational web after extracting entities from 1000+ lists in Witcher books using Python (Selenium, spaCy NER and Pandas).

Education

Master of Science (M.Sc.) – Management and Engineering CAME Oct 2022 – Mar 2026
RWTH Aachen University *Aachen, Germany*
CGPA: **1.7**
Relevant Courses: Strategy Management, AI & Data Analytics, Finance Management, Marketing, Management & Engineering Perspectives

Bachelor of Technology (B.Tech.) – Mechanical Engineering Aug 2016 – Oct 2020
KJ Somaiya College of Engineering (Mumbai University) *Mumbai, India*
CGPA: **1.4** (German grade equivalent)
Bachelor's Thesis: Amphibious Garbage Collector Robot (Designing and Simulation)

Technical Skills

- **Analysis & Visualization:** Excel, Power BI, Tableau, Think Cell
- **Programming:** Python (Pandas, Numpy, Matplotlib), R (Tidyverse, base R), MySQL, MATLAB
- **AI Tools:** Claude, Gemini, Perplexity, ChatGPT (automating-reporting, research synthesis, workflow speed)
- **Project Management & CRM:** SAP, S/4HANA, Confluence, Jira, MiroBoard, Agile/Scrum
- **Productivity & Technical:** PowerPoint, SharePoint, LaTeX, SolidWorks, AutoCAD

Languages

English (C1 Fluent) | **Deutsch** (B1 Intermediate) | Marathi (Native) | Hindi (Native)

Courses and Certificates

BCG	Strategy Consulting Job Simulation
McKinsey & Co.	McKinsey Forward Program
PWC	Power BI Analysis
Fraunhofer LMS	E-Learning Scrum | E-Learning Kanban
Google	Google AI Essentials | Prompt Engineering
IBM	Python for Data Science | Databases & SQL for Data Science

Hobbies

Film Critique | Team Sports (Football) | Drawing/Painting | Singing | Volunteering Cook at Tivoli Aachen